

Ito Calculus

Brownian motion B_t

Quadratic Variation $(dB_t)^2 = dt$

(Simple) Ito's Lemma $f(B_t)$

$$df = f'(B_t) dB_t + \frac{1}{2} f''(B_t) dt$$

Taylor Expansion

$f(t, x)$

$df = ?$

$$f(t + \Delta t, x + \Delta x) = f(t, x) + \frac{\partial f}{\partial t} \Delta t + \frac{\partial f}{\partial x} \Delta x$$

$$+ \frac{1}{2} \left(\frac{\partial^2 f}{\partial t^2} (\Delta t)^2 + 2 \frac{\partial^2 f}{\partial t \partial x} (\Delta t)(\Delta x) + \frac{\partial^2 f}{\partial x^2} (\Delta x)^2 \right)$$

$$\Rightarrow \underline{f(t + dt, B_t + dB_t) - f(t, B_t)} = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial B_t} dB_t + \frac{1}{2} \frac{\partial^2 f}{(\partial B_t)^2} dt$$

$$= \left(\frac{\partial f}{\partial t} + \frac{\partial^2 f}{2 (\partial B_t)^2} \right) dt + \frac{\partial f}{\partial B_t} (dB_t)$$

Consider a stochastic Process,

$$dX_t = \underbrace{\mu dt}_{\text{drift term}} + \sigma dB_t$$

$$X_t = \sigma B_t + \mu t$$

if μ & σ const.

$$X_t = \mu t + \sigma B_t$$

$$\frac{d(B_t)^2}{dt} = dt$$

Theorem (Ito's Lemma)

f is a smooth function

$$X_t: dX_t = \mu dt + \sigma dB_t$$

$$\Rightarrow df(t, x_t) = \left(\frac{\partial f}{\partial t} + \mu \frac{\partial f}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2 f}{\partial x^2} \right) dt$$

$$dX_t = \mu dt + \sigma dB_t + \frac{\partial f}{\partial x} dB_t$$

$$\Rightarrow f(t, x) = x^2 \quad (x_t)$$

$$f(t, B_t) = ?$$

$$\mu = 0$$

$$\sigma = 1$$

$$\Rightarrow X_t: B_t$$

$$df(B_t) = dt + 2B_t dB_t$$

Ex 2 $f(t, x) = e^{\mu t + \sigma x}$

$$df(B_t) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dx + \frac{1}{2} \frac{\partial^2 f}{(\partial x)^2} dt$$

$$df(B_t) = e^{\mu t + \sigma x} \left(\mu dt + \sigma dx + \frac{\sigma^2}{2} dt \right)$$

$\Rightarrow$ Want to model Stock Price S_t using general Brownian motion.

$$\frac{dS_t}{S_t} = \sigma dB_t$$

Q) $\exists S_t = e^{\sigma B_t}$

$$\Rightarrow dS_t = \underbrace{\frac{1}{2} \sigma^2 dt}_{\text{Drift } S_t} + \sigma dB_t$$

$\Rightarrow$ For drift $\neq$ zero,

$$S_t = e^{\sigma B_t - \frac{1}{2} \sigma^2 dt}$$

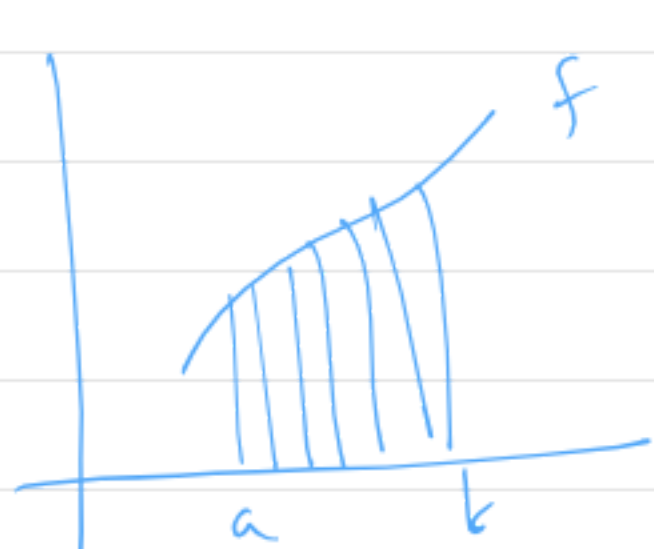
Def We define

$$F(t, B_t) = \int f(t, B_t) dB_t + \int g(t, B_t) dt$$

if

$$dF = f dB_t + g dt$$

$\Rightarrow$ Is there a Riemannian Sum type desc.?

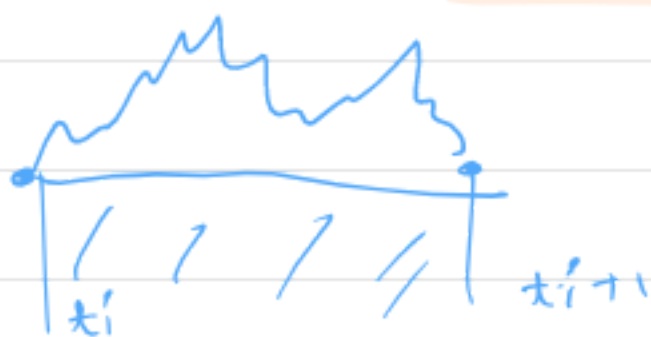


$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n f\left(\frac{ib}{n}\right)$$

in limit, we take middle point, which is not possible here due to quadratic variance.

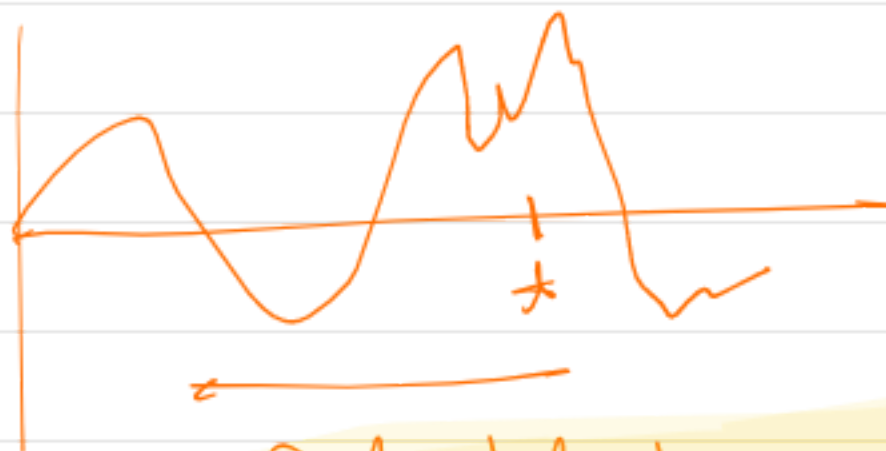
$\Rightarrow$ Ito interval is the limit of Riemannian Sum where we always choose left most point at each interval.

$\Rightarrow$ It works best, because we make decision at starting of interval (no future insights)



Adapted Process -

$\Delta(t)$ is adapted to X_t if $\forall t \geq 0$, Δt only depends on $X_0 \sim X_t$



Only depends on past

prefix sum $\Rightarrow$

	1	0	2	1	1
$\Rightarrow$	1	1	3	4	5

Ex. X_t is adapted to X_t

$Y_t = X_{t+1}$ is not adapted to X_t .

$\Delta t = \min\{X_t, c\}$ is adapted to X_t

$T > 0$ $\Delta t = \max_{t < s < T} X_s$ not adapted

$$B(t) \sim N(0, t)$$

$$\underline{X(t)} = \underline{\nabla B(t)} \sim N(0, \underline{\sigma^2 t})$$

$$= \int \nabla dB_t \leftarrow$$

$\Rightarrow$ Take all small steps of B , and scale them by ∇ .

Theorem

If Δt is process dependent only on t ,

then,

$$X(t) = \int \Delta(t) dB_t$$

has Normal distribution all the time

$\hookrightarrow$ sum of Normal variables

Ito's Isometry

$\Delta(t)$: adapted to B_t

$$\mathbb{E} \left[\left(\int_0^t \Delta(s) dB_s \right)^2 \right] = \mathbb{E} \left[\int_0^t \Delta^2(s) ds \right]$$

For $s=1$

$$\mathbb{E} \left[(B_t)^2 \right] = t \quad (dB_t)^2 = dt \quad (\text{Quadratic Variance})$$

When is Ito Integral a Martingale?
(Fair Game)

Thm If $g(t, B_t)$ is adapted to B_t , then

$\int g(t, B_t) dB_t$ is a martingale as long as g is "reasonable".

$$\iint g^2(t, B_t) dt dB_t < \infty$$

$$dX_t = \underbrace{\mu(t, B_t) dt}_{\text{Drift term}} + \sigma(t, B_t) dB_t$$

X_t is a martingale if $\mu = 0$ always.

Change of Measure (Girsanov's Thm)

Q) B_t : Brownian Motion with No Drift
 $\tilde{B}_t$: " " " Drift μ



Can we switch b/w the two prob. dist. by a change of measure?

$\Rightarrow P(\omega)$ given by $B_t \leftarrow$
 $\tilde{P}(\omega)$ given by $\tilde{B}_t \leftarrow$

$\exists Z = Z(\omega)$ s.t. $P(\omega) = Z(\omega) \tilde{P}(\omega)$?

(Ω, P) : Prob. dist.
 $(\Omega, \tilde{P})$: "

- Ω = sample space (all possible outcomes).
- P = a probability measure/distribution.
- $\tilde{P}$ = another probability measure/distribution on the same Ω .

For example, if $\Omega = \{H, T\}$:

- Under P : $P(H) = 0.5$, $P(T) = 0.5$.
- Under $\tilde{P}$: $\tilde{P}(H) = 0.7$, $\tilde{P}(T) = 0.3$.

Def

P and $\tilde{P}$ are eq. if

$$P(A) > 0 \iff \tilde{P}(A) > 0 \\ \forall A \subseteq \Omega$$

$$P = \{1, 2, 3\}$$

$$\rightarrow \frac{1}{3} \quad \frac{1}{3} \quad \frac{1}{3}$$

$\downarrow \downarrow$
 $\neq$

$\searrow$
 $\times$

$$\tilde{P} = \{1, 2, 3\}$$

$$\rightarrow \frac{2}{3} \quad \frac{1}{6} \quad \frac{1}{6}$$

$$\downarrow \downarrow$$

$$\rightarrow \frac{2}{3} \quad \frac{1}{3} \quad 0$$

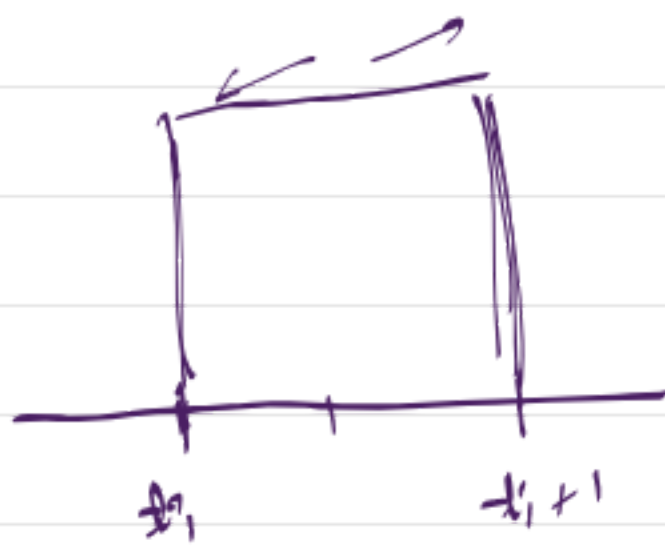
Thm $\exists Z$ s.t. $P(\omega) = Z(\omega) \tilde{P}(\omega)$

iff P and $\tilde{P}$ are eq-

$\Rightarrow$ Random - Nikodym derivative

$\Rightarrow$ If $P(\omega)$ and $\tilde{P}(\omega)$ are eq-

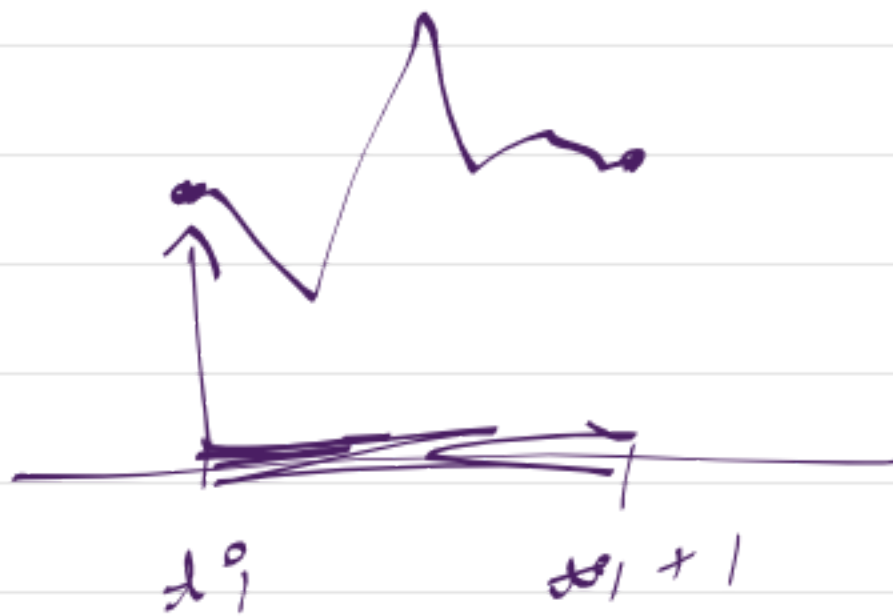
$$Z(\omega) = \frac{d\tilde{P}}{dP}(\omega) = e^{-n\omega(x) - n^2T/2}$$



$$h \Rightarrow (x_{i+1})$$

$$h \Rightarrow (x_i)$$

$$h \Rightarrow \left(\frac{x_{i+1} + x_i}{2} \right) \quad \checkmark$$



1s

1.5s

2s